

Homework 2

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Problem 1

Part a

$$E(Y_t) = E(aX_{t-\ell} + W_t) = aE(X_{t-\ell}) + E(W_t) = a \cdot 0 + 0 = 0.$$

$$\text{Var}(Y_t) = \text{Var}(aX_{t-\ell} + W_t) = a^2\text{Var}(X_{t-\ell}) + \text{Var}(W_t) = a^2\sigma_X^2 + \sigma_W^2,$$

Since X_t is stationary, γ_x only depends on the lag and not time, so the ACF of Y_t also will only depend on the lag and not time because W_t is also stationary.

Part b

$$\text{Cov}(Y_t, X_s) = E[Y_t X_s] - E(Y_t)E(X_s).$$

$$\text{Cov}(Y_t, X_s) = E[(aX_{t-\ell} + W_t)X_s] = aE[X_{t-\ell}X_s] + E[W_tX_s].$$

$$\text{Cov}(Y_t, X_s) = aE[X_{t-\ell}X_s] = a\gamma_X(t - \ell - s),$$

Part c

$$\rho_{Y_t, X_s} = \frac{\text{Cov}(Y_t, X_s)}{\sqrt{\text{Var}(Y_t)} \cdot \sqrt{\text{Var}(X_s)}}.$$

$$\text{Cov}(Y_t, X_s) = a\gamma_X(t - \ell - s),$$

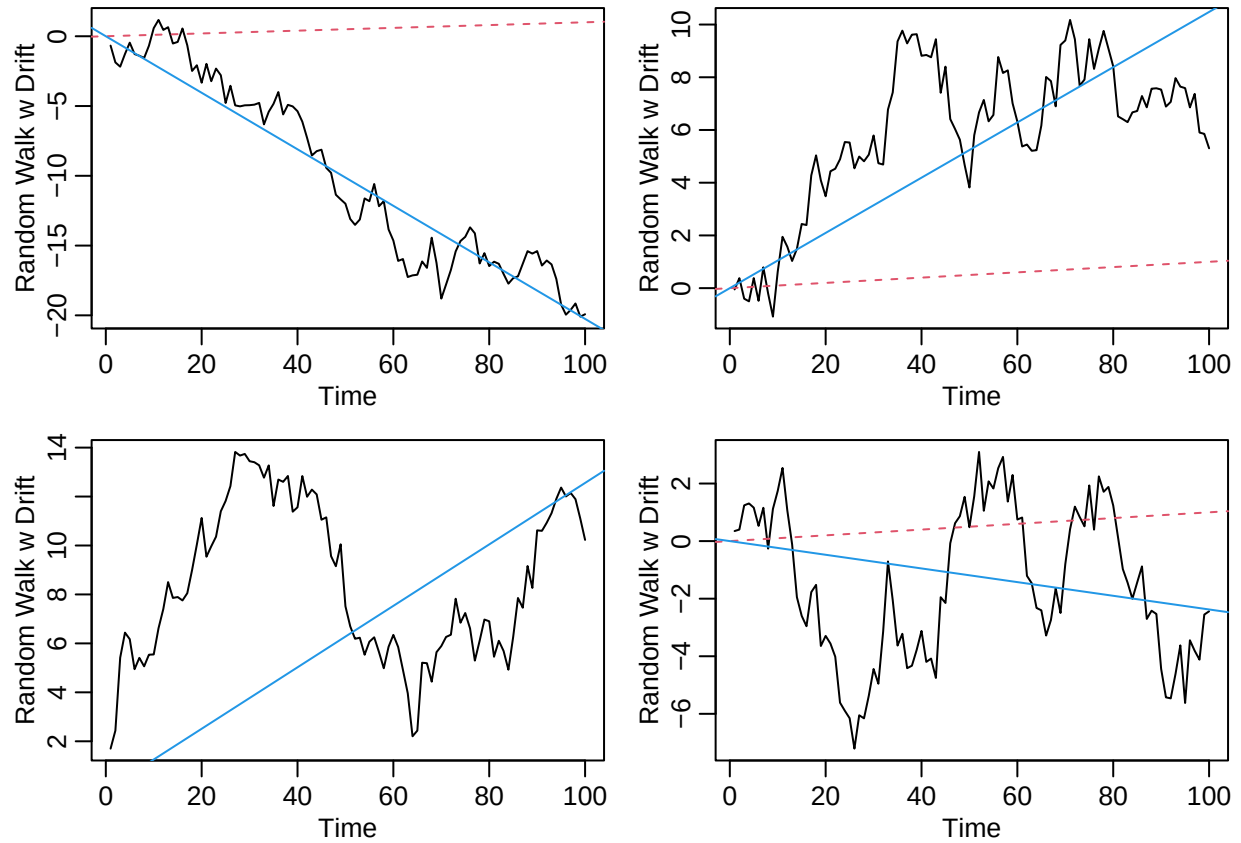
$$\text{Var}(Y_t) = a^2\sigma_X^2 + \sigma_W^2 \quad \text{and} \quad \text{Var}(X_s) = \sigma_X^2.$$

$$\rho_{Y_t, X_s} = \frac{a\gamma_X(t - \ell - s)}{\sqrt{a^2\sigma_X^2 + \sigma_W^2} \cdot \sigma_X}.$$

Problem 2

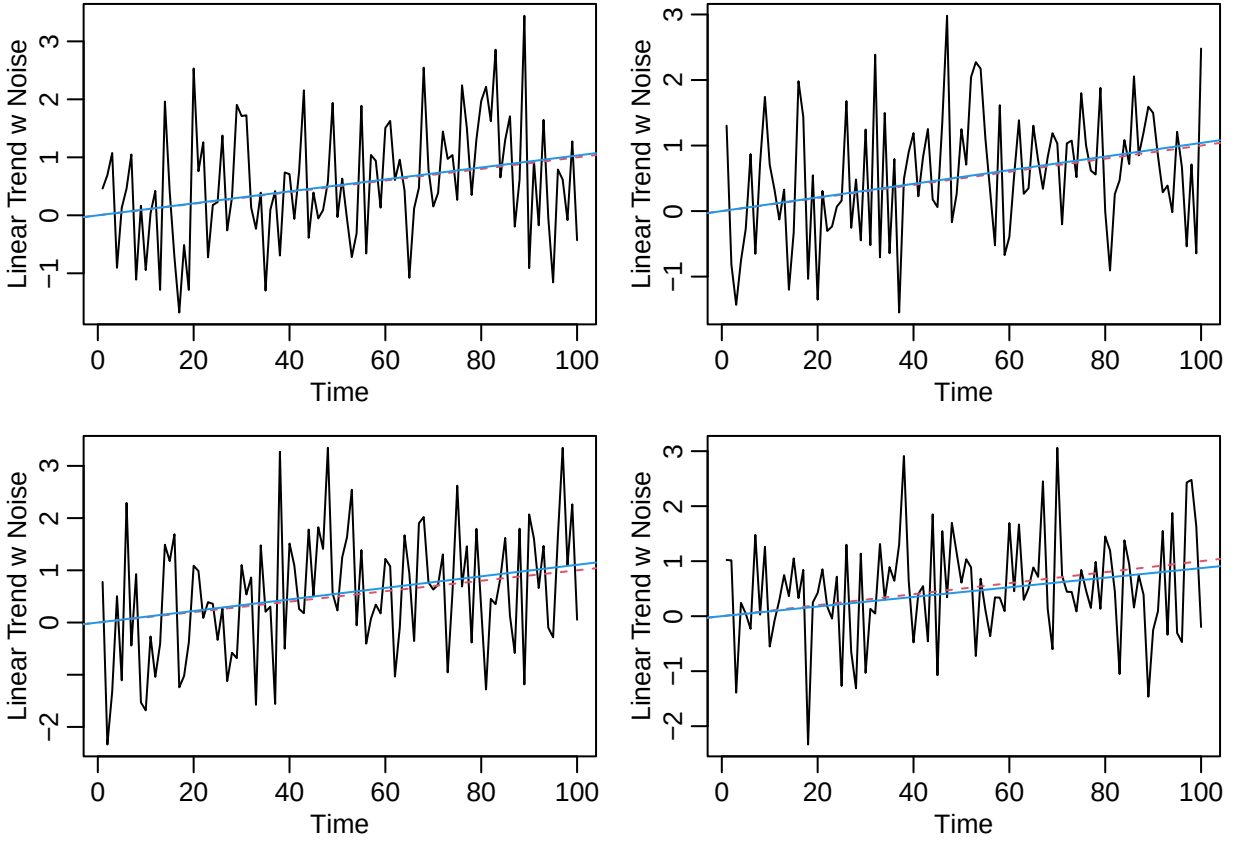
Part a

Random Walks with drift:



Part b

Linear Trend Series with Noise:



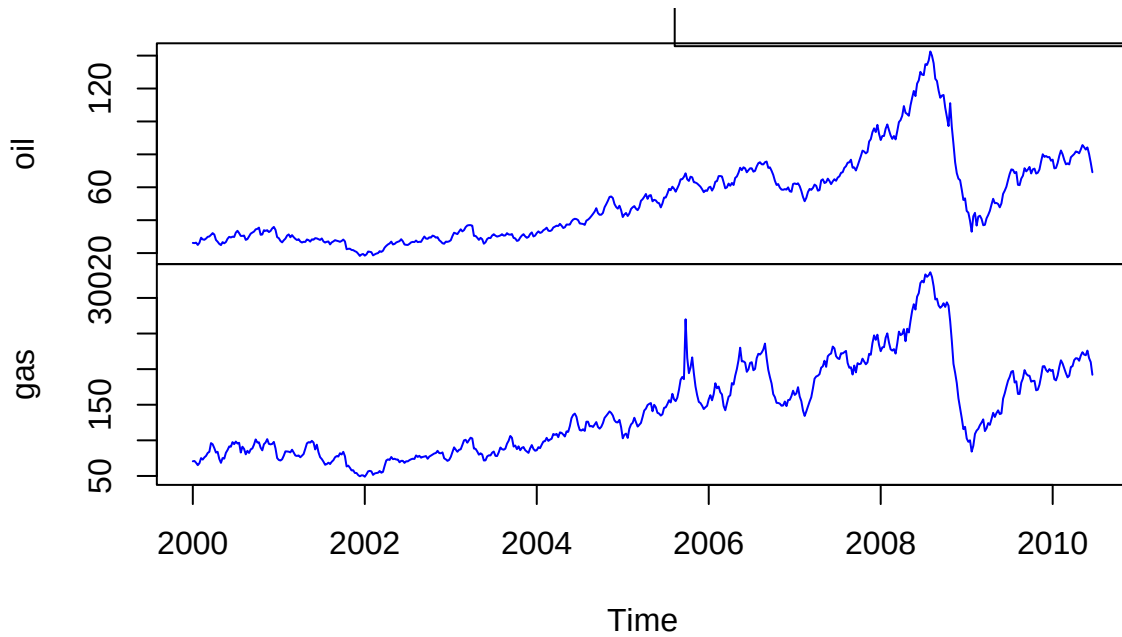
Part c

Despite these series having the same expected value and variance in generating each data point, the variance of the random walk as a whole is much larger than that of the linear trend with noise because each observation of the random walk is dependent on the previous observation. Neither of these series are stationary because the mean increases with time but the random walk wouldn't be stationary even if detrended because the variance also increases with time whereas the linear trend series with noise would be.

Problem 3

Part a

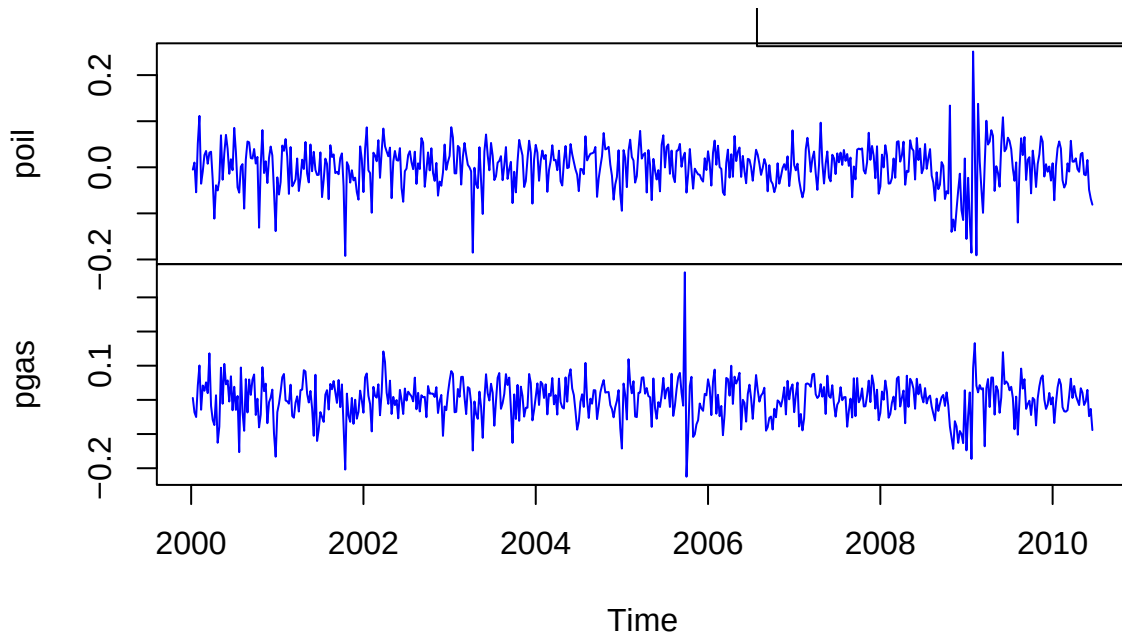
Oil and Gas Prices



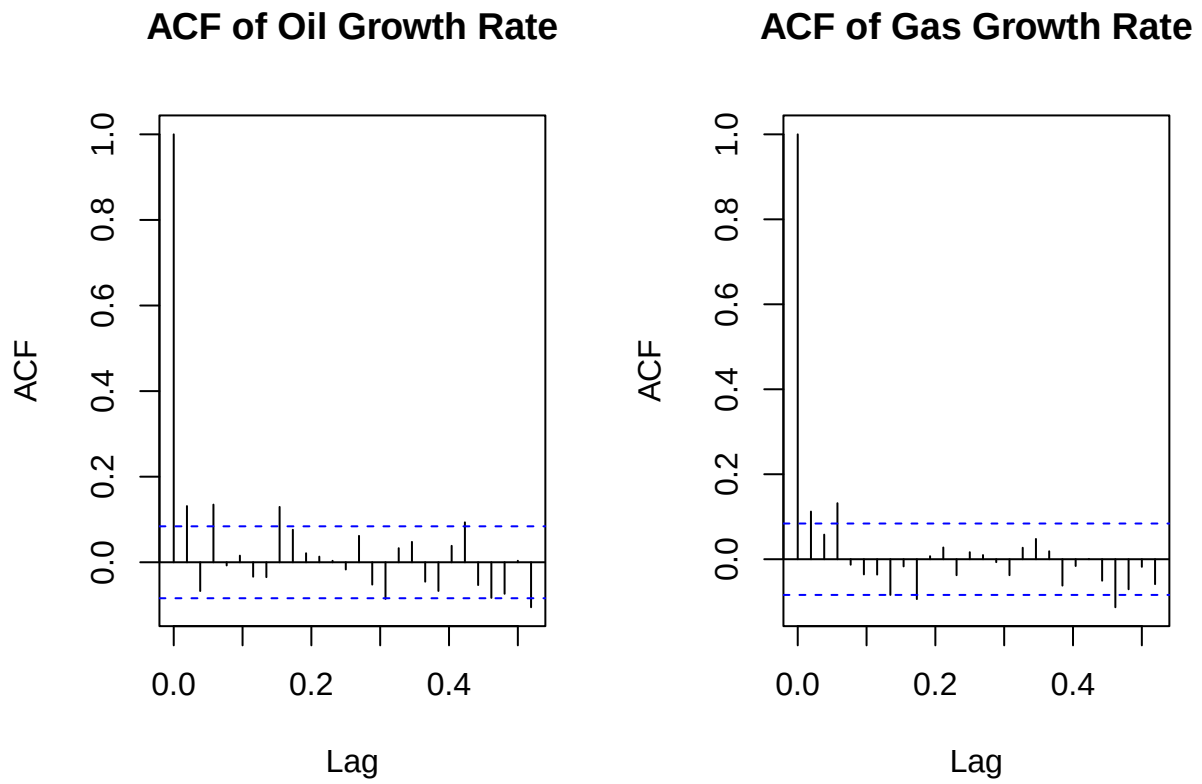
These series best resemble a random walk with drift. This leads me to believe they are not stationary because both the mean and the variance appear to increase with time.

Parts b and c

Transformed Oil and Gas Growth Rates



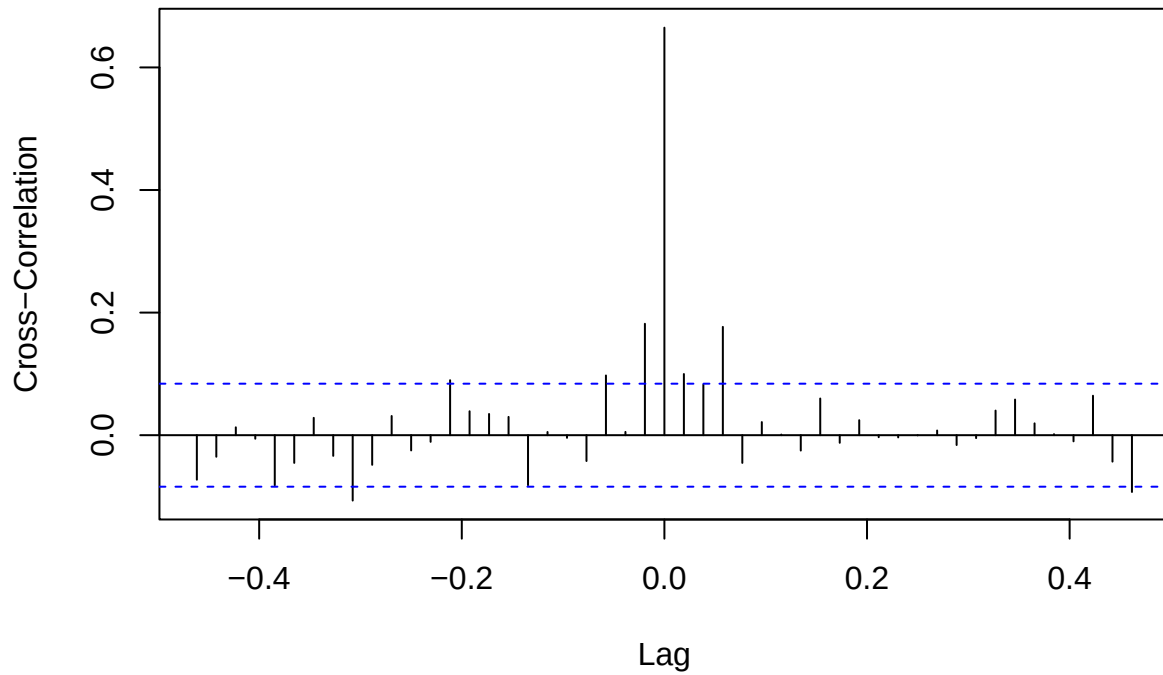
This plot is useful because it makes it easier to see the growth of the prices of oil and gas than the plots from part a. In the plots from part a, changes towards the beginning of the series are harder to see because they are much smaller in scale than those of the later portion of the series. This also provides a stationary representation of the changes in oil and gas prices whereas the plots from part a were not stationary.



The ACFs look pretty much as one would want them to for a stationary time series because the values after $h=0$ are comparatively very small, with none exceeding 0.15.

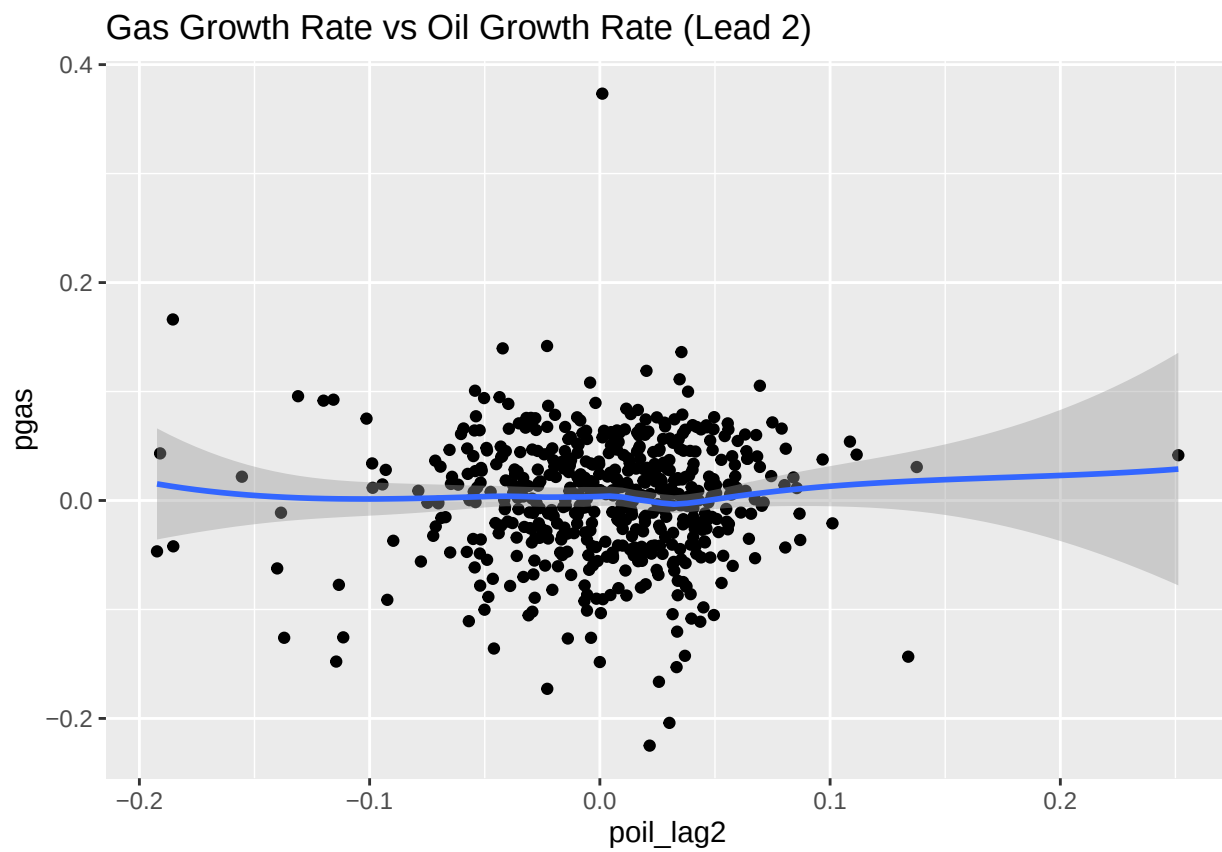
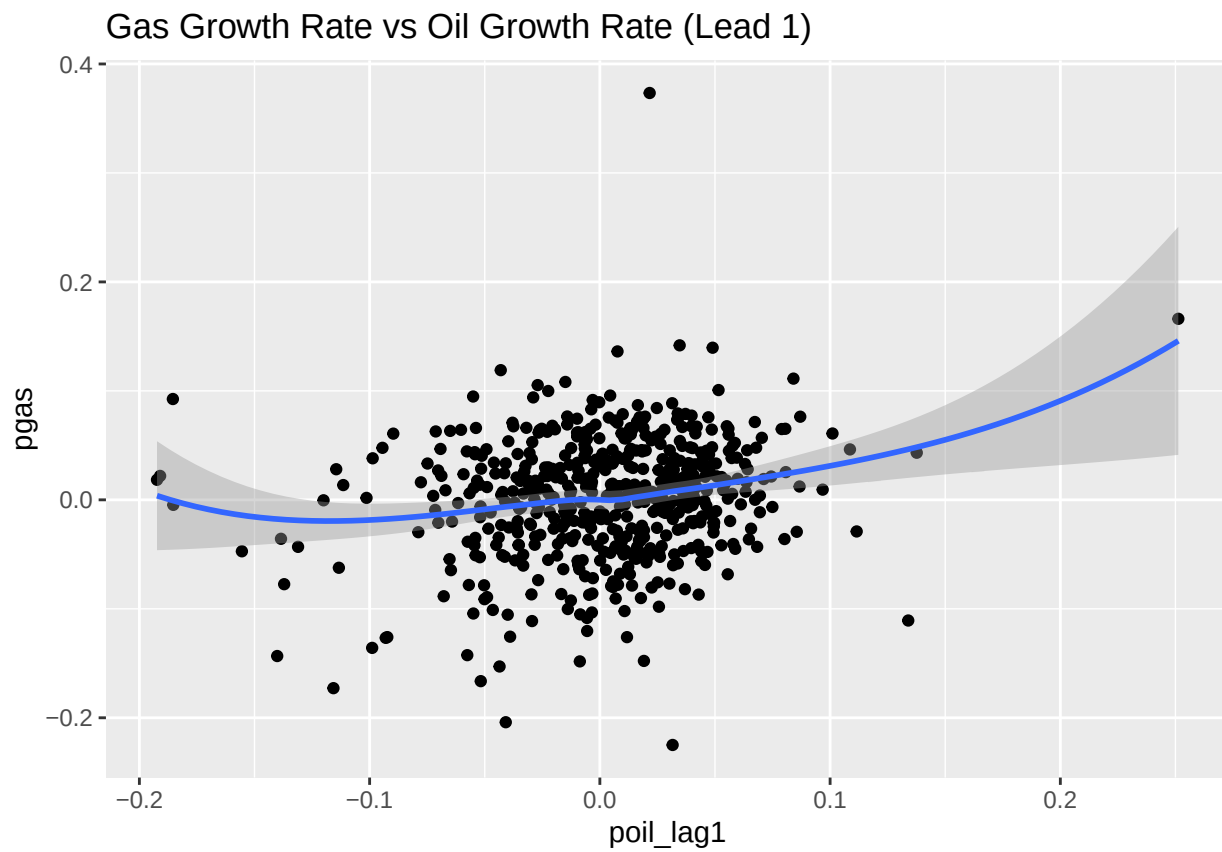
Part d

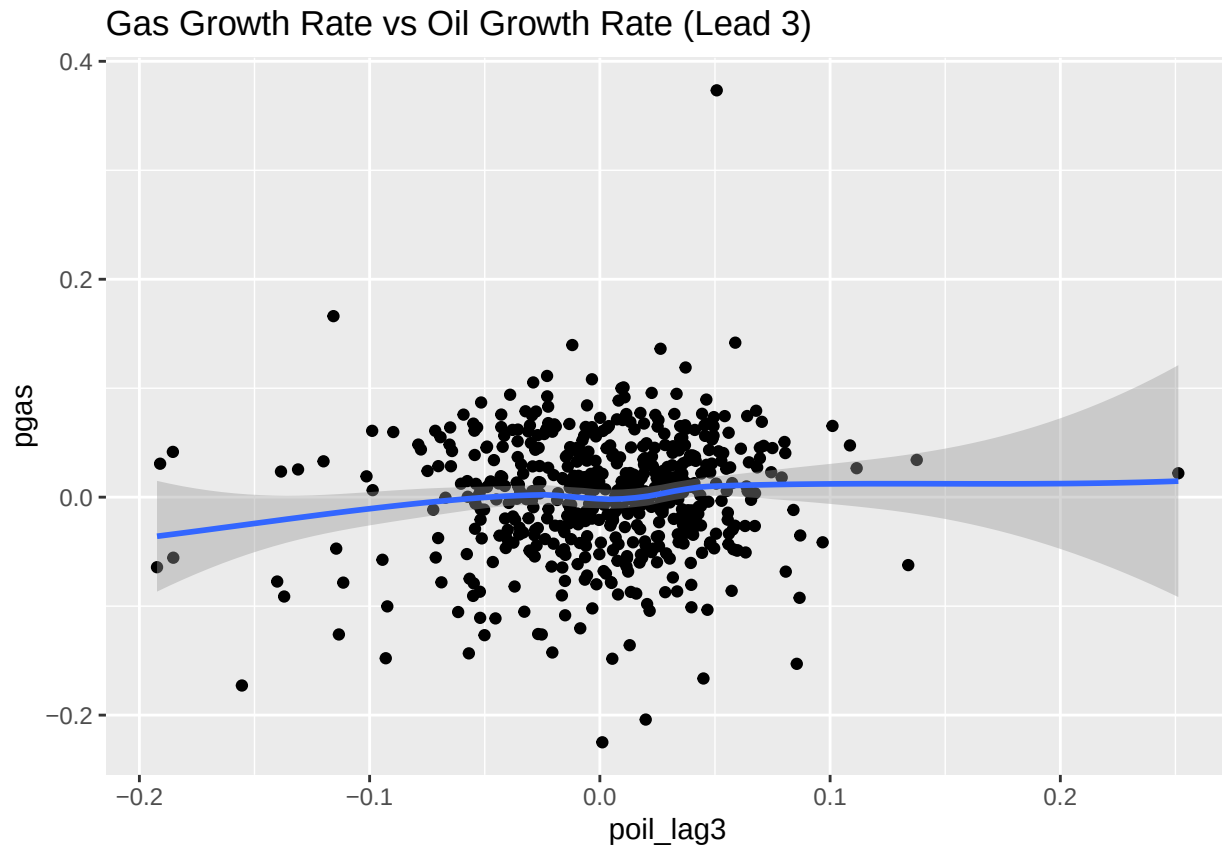
CCF of Oil and Gas Growth Rates



The most significant cross correlation by far is at $h=0$, which one might expect knowing that gasoline is made from crude oil. As mentioned in the problem we see a slight correlation in first the few observations when oil leads gas because the gas is produced from the oil. The observation right before $h=0$ appears barely significant likely just because the price growth has some very small dependence on time but this is acceptable for real data given that none of the other observations appear to have significant cross correlation.

Part e



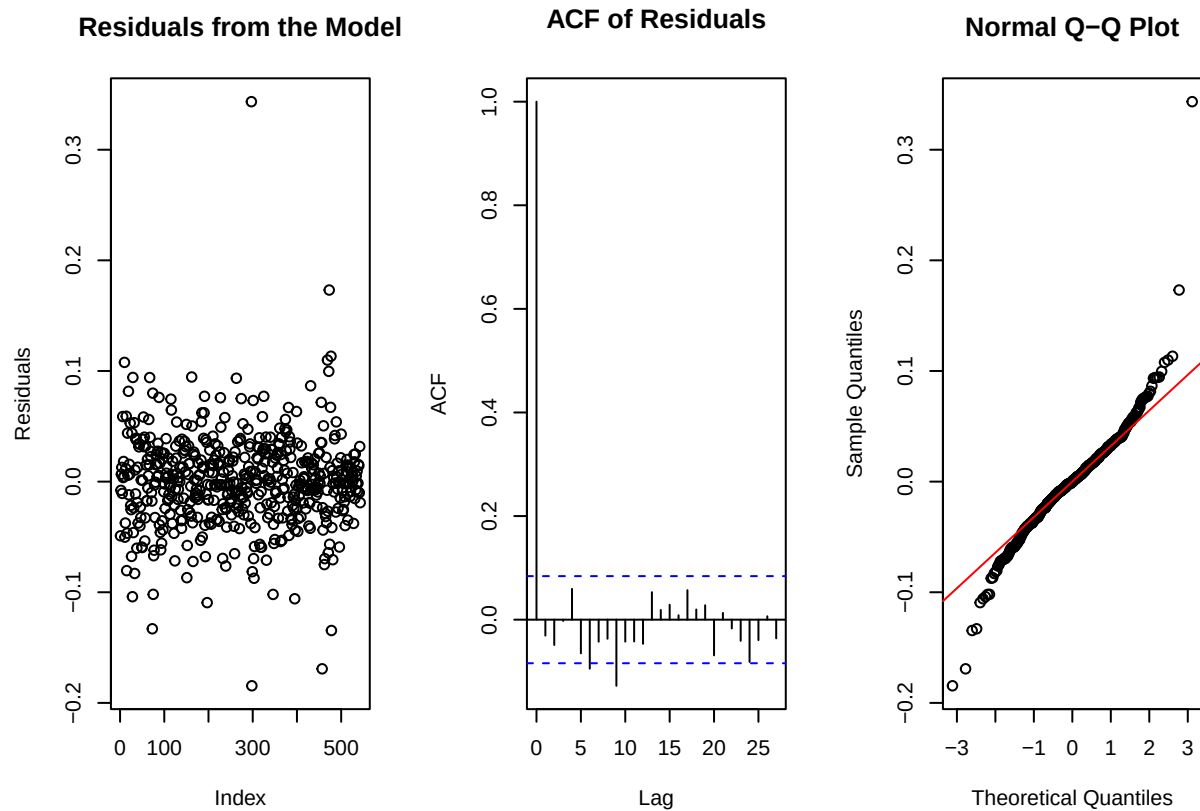


In each of these plots, there is one outlier on the x axis (lag 1, 2, and 3 of oil price growth rate) and one on the y axis (gasoline price growth rate). The y axis outlier has some influence on the correlation, though not a ton because the x axis values are pretty close to zero for that outlier in all 3 plots. The x axis outlier seems to have an outsized influence on the nonparametric smoother in the lag 1 plot. There is not a very clear linear relationship for any of them as the nonparametric smoothers are fairly flat apart from the part due to the influence of the x axis outlier in the lag 1 plot.

Part f

```
##
## Call:
## lm(formula = pgas ~ poil + poilL + indi, data = mess)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.18451 -0.02161 -0.00038  0.02176  0.34342
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.006445   0.003464  -1.860  0.06338 .
##      poil      0.683127   0.058369  11.704 < 2e-16 ***
##     poilL      0.111927   0.038554   2.903  0.00385 **
##      indi      0.012368   0.005516   2.242  0.02534 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 0.04169 on 539 degrees of freedom
## Multiple R-squared:  0.4563, Adjusted R-squared:  0.4532
## F-statistic: 150.8 on 3 and 539 DF,  p-value: < 2.2e-16
```



- The oil price growth, the lag 1 of oil price growth, and the binary indicator of positive oil price growth all have positive betas and are all significant at the $p < 0.05$ level.
- When the growth of the price of oil is negative, $G_t = -0.006445 + 0.683127O_t + 0.111927O_{t-1}$. When the growth of the price of oil is positive, $G_t = -0.006445 + 0.012368 + 0.683127O_t + 0.111927O_{t-1}$. This suggests that the price of gas responds more quickly to rising oil prices than falling oil prices.
- The residuals appear to be close to normally distributed with equal variance, the ACF looks as it should for a stationary series with no values other than at $h=0$ with an absolute value above 0.1, and the normal QQ plot shows that the distribution is slightly wider than normal but nothing extreme.

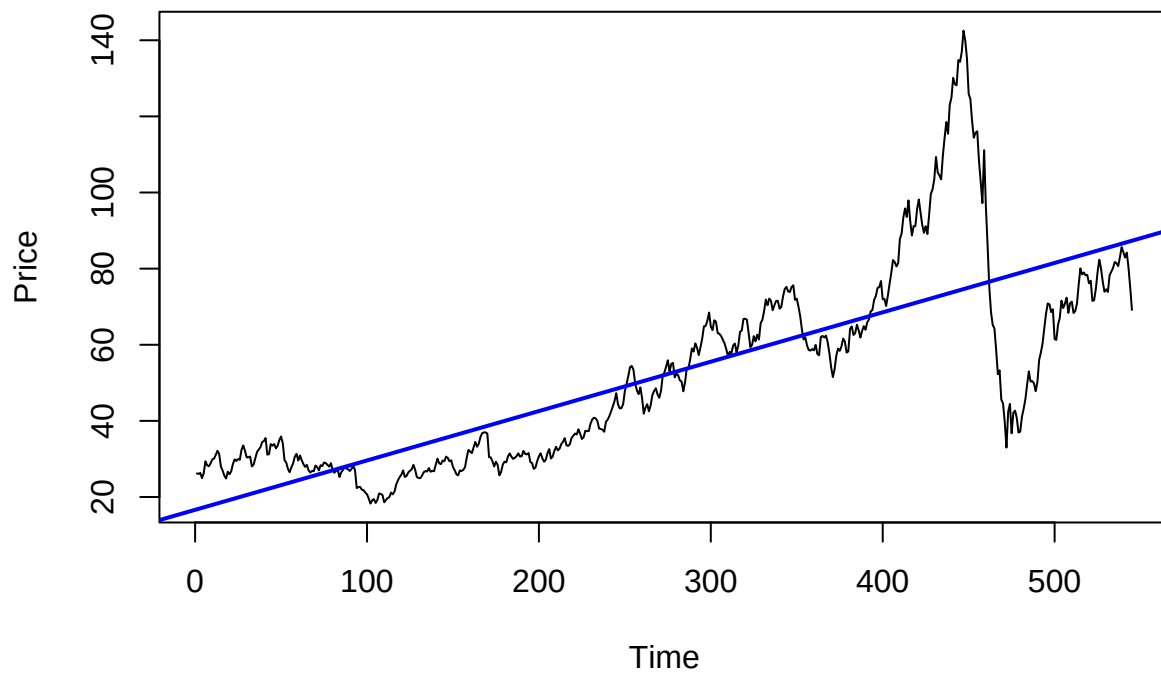
Problem 4

Part a

```
##
## Call:
## lm(formula = oil ~ time)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -44.889 -9.079 -2.731 7.389 67.894
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 16.63690    1.36760   12.16  <2e-16 ***
## time         0.12973    0.00434   29.89  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.94 on 543 degrees of freedom
## Multiple R-squared:  0.622, Adjusted R-squared:  0.6213
## F-statistic: 893.4 on 1 and 543 DF, p-value: < 2.2e-16
```

Oil Prices with Linear Regression Line



The coefficients of time and intercept are both statistically significant at the $p < 0.001$ level.

Part b

```
## F-statistic: 893.3612
## p-value for delta_1: 8.670476e-117
```

Part c

```
## Proportion of p-values < 0.05: 0.956
```

While the proportion of p values below 0.05 is very high (0.956), it not being closer to 1 suggests that the extremely small p value from part b ($8.670476 \times 10^{-117}$) does not always translate when the random walk is used instead of just a linear model.